

The Trojan Pop-Up

4dw@r3, They/She

ROMHACK CAMP

Are all digital adverts Adware?

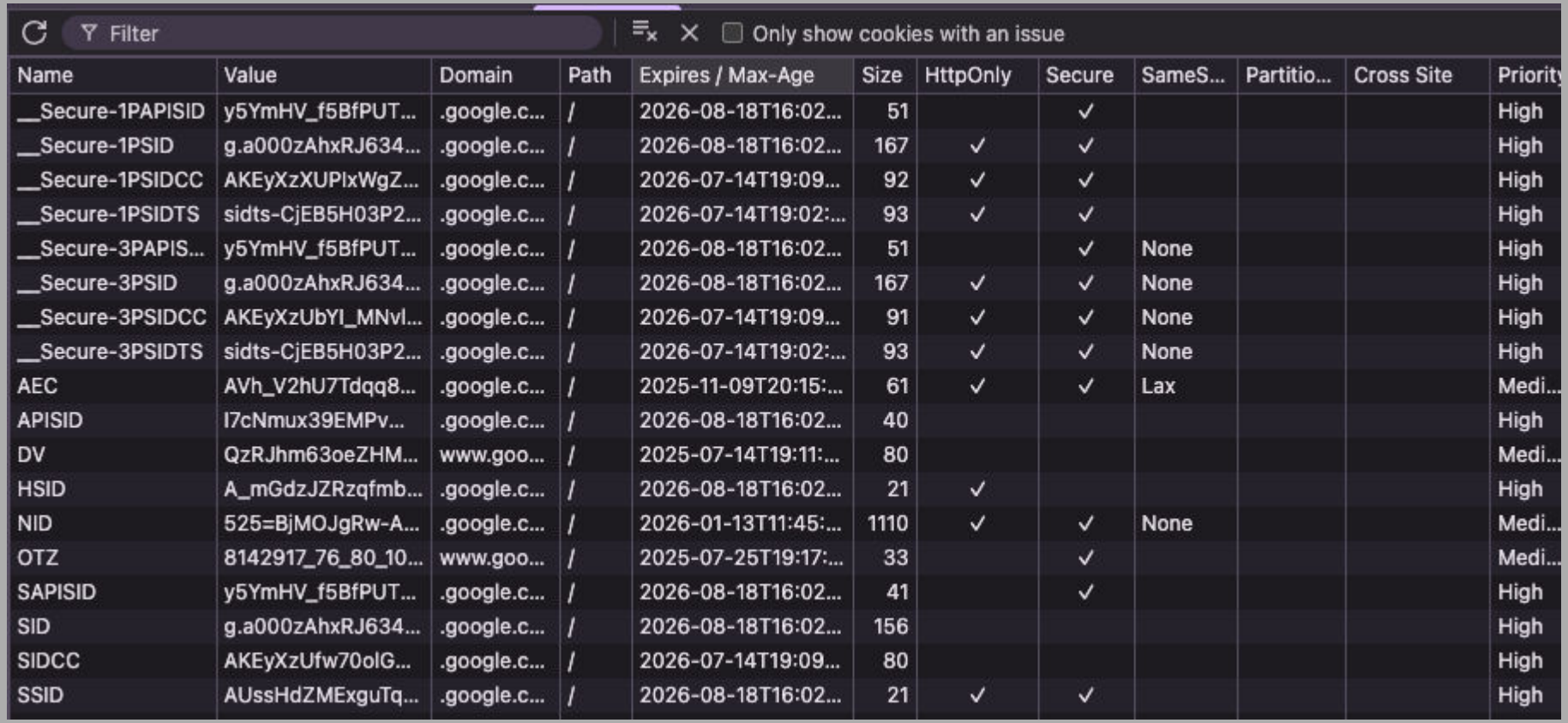


How do ad agencies consume most of their data?

Cookies



What are cookies?



Filter

Only show cookies with an issue

Name	Value	Domain	Path	Expires / Max-Age	Size	HttpOnly	Secure	SameS...	Partitio...	Cross Site	Priority
__Secure-1PAPISID	y5YmHV_f5BfPUT...	.google.c...	/	2026-08-18T16:02...	51		✓				High
__Secure-1PSID	g.a000zAhxRJ634...	.google.c...	/	2026-08-18T16:02...	167	✓	✓				High
__Secure-1PSIDCC	AKEyXzXUPlxWgZ...	.google.c...	/	2026-07-14T19:09...	92	✓	✓				High
__Secure-1PSIDTS	sidts-CjEB5H03P2...	.google.c...	/	2026-07-14T19:02:...	93	✓	✓				High
__Secure-3PAPIS...	y5YmHV_f5BfPUT...	.google.c...	/	2026-08-18T16:02...	51		✓	None			High
__Secure-3PSID	g.a000zAhxRJ634...	.google.c...	/	2026-08-18T16:02...	167	✓	✓	None			High
__Secure-3PSIDCC	AKEyXzUbYI_MNvl...	.google.c...	/	2026-07-14T19:09...	91	✓	✓	None			High
__Secure-3PSIDTS	sidts-CjEB5H03P2...	.google.c...	/	2026-07-14T19:02:...	93	✓	✓	None			High
AEC	AVh_V2hU7Tdqq8...	.google.c...	/	2025-11-09T20:15:...	61	✓	✓	Lax			Medi...
APISID	I7cNmux39EMPv...	.google.c...	/	2026-08-18T16:02...	40						High
DV	QzRJhm63oeZHM...	www.goo...	/	2025-07-14T19:11:...	80						Medi...
HSID	A_mGdzJZRzqfmb...	.google.c...	/	2026-08-18T16:02...	21	✓					High
NID	525=BjMOJgRw-A...	.google.c...	/	2026-01-13T11:45:...	1110	✓	✓	None			Medi...
OTZ	8142917_76_80_10...	www.goo...	/	2025-07-25T19:17:...	33		✓				Medi...
SAPISID	y5YmHV_f5BfPUT...	.google.c...	/	2026-08-18T16:02...	41		✓				High
SID	g.a000zAhxRJ634...	.google.c...	/	2026-08-18T16:02...	156						High
SIDCC	AKEyXzUfw70olG...	.google.c...	/	2026-07-14T19:09...	80						High
SSID	AUssHdZMExguTq...	.google.c...	/	2026-08-18T16:02...	21	✓	✓				High

What is a cookie?

A small text file often with an encrypted string in it



Used for innocent things like remembering you like dark mode

But also used to track your location, SSIDs accessed, and web history



Also used as session tokens, vulnerable to theft

Marketing firm 101

Marketing firms/Ad Agencies: place and track ads

Advertisers: pay for ads

Agency Revenue

Agencies charge a percentage of ad spend for creating/ placing ads

Most ad agency revenue is from secondary analytics services

The death of cookies?

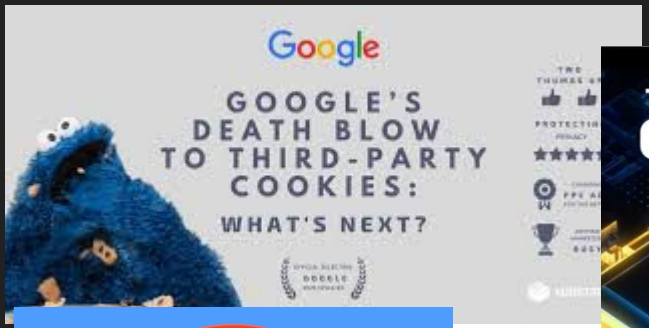


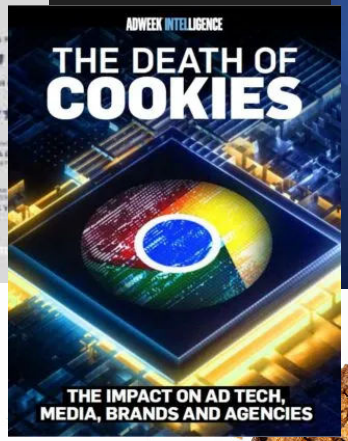
The death of cookies?

GDPR specifically targeted cookies, and third party data exchange

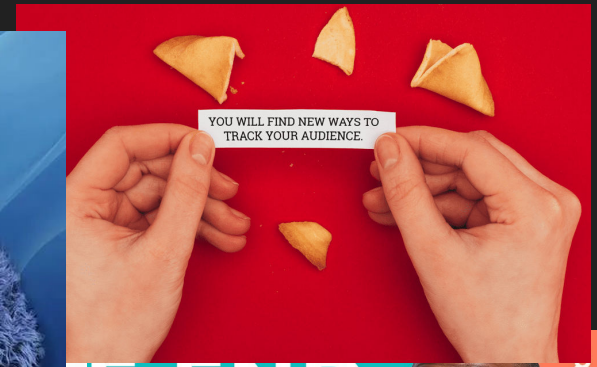
Apple and others started limiting compatibility with cookies for user privacy reasons

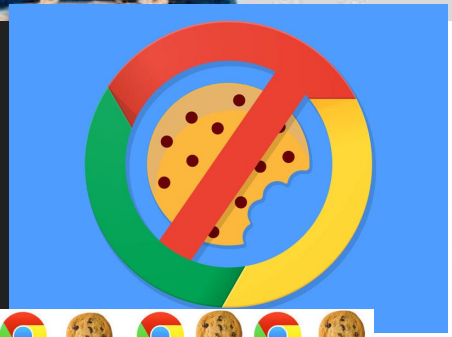
Google launched an initiative to kill third-party cookies in Chrome by 2020, calling it: Privacy Sandbox (Google Analytics 4)


Google
GOOGLE'S DEATH BLOW TO THIRD-PARTY COOKIES:
WHAT'S NEXT?


ADWEEK INTELLIGENCE
THE DEATH OF COOKIES
THE IMPACT ON AD TECH, MEDIA, BRANDS AND AGENCIES


RIP


YOU WILL FIND NEW WAYS TO TRACK YOUR AUDIENCE.






THE DEATH OF COOKIES


E END
RD-PARTY
KIES


Saying goodbye to third-party cookies in 2024

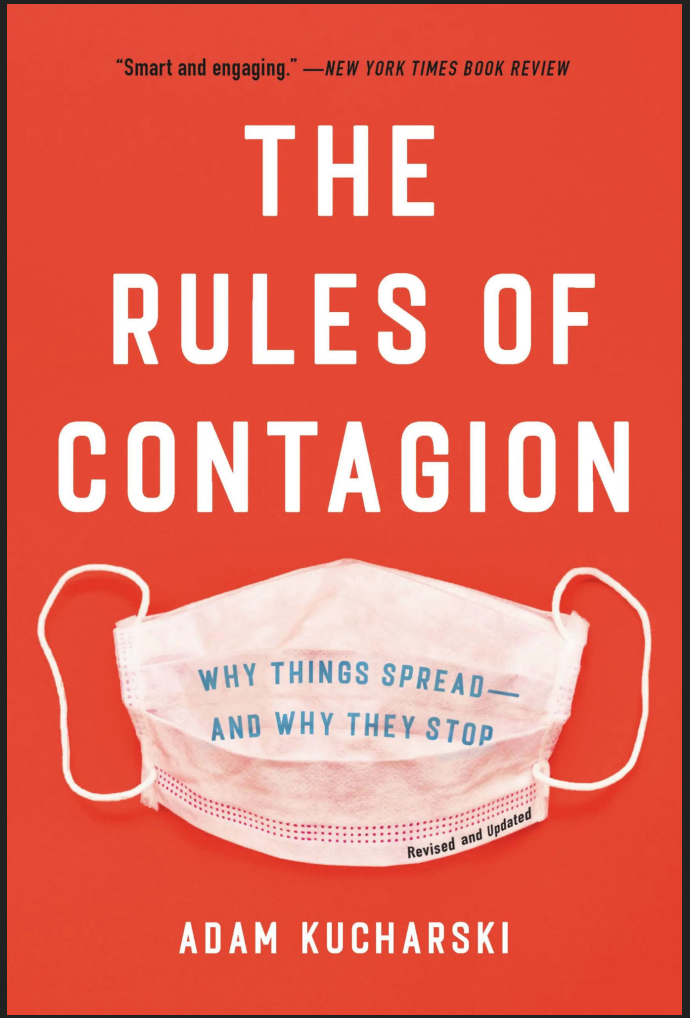

What Does the Death of Third-Party Cookies Mean for Online Advertising?

Dark money panic

State-actors, and superPACs track engagement with misinformation

Studying: keywords, engagement, disengagement, and

Post release **spread**



Privacy Advocates Shrug

Eliminating cookies decreases access to information for 3rd parties



It consolidates access and empowers Google’s monopoly

Doesn’t seem like the privacy *win* Google promised

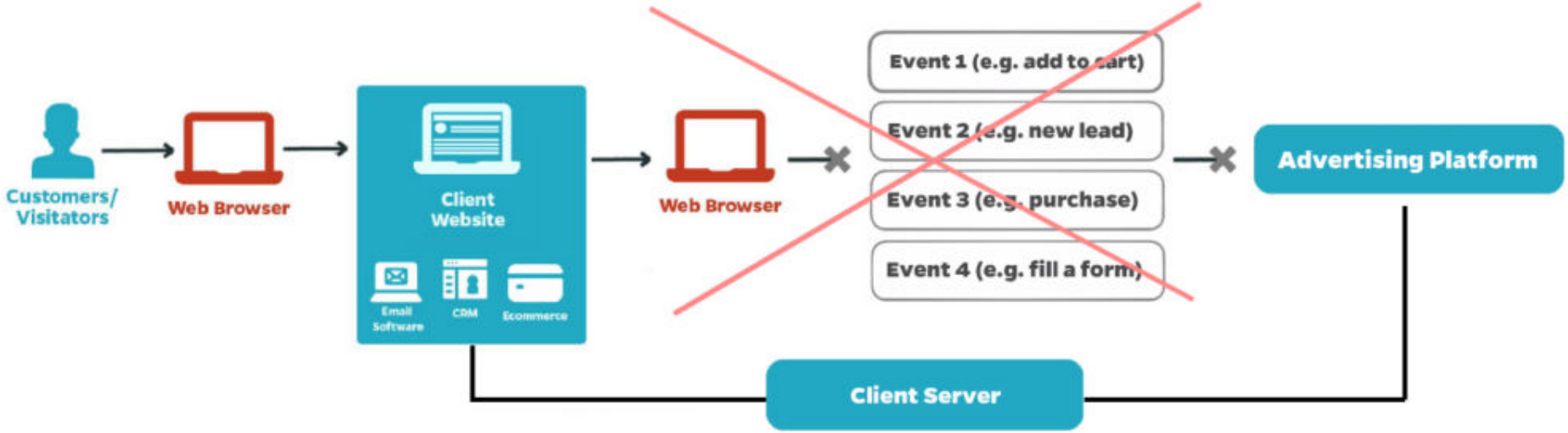
Ad agencies to the rescue!

The most brilliant strategic minds in marketing put their thinking caps on and came up with a ‘new’ type of tech to solve this :

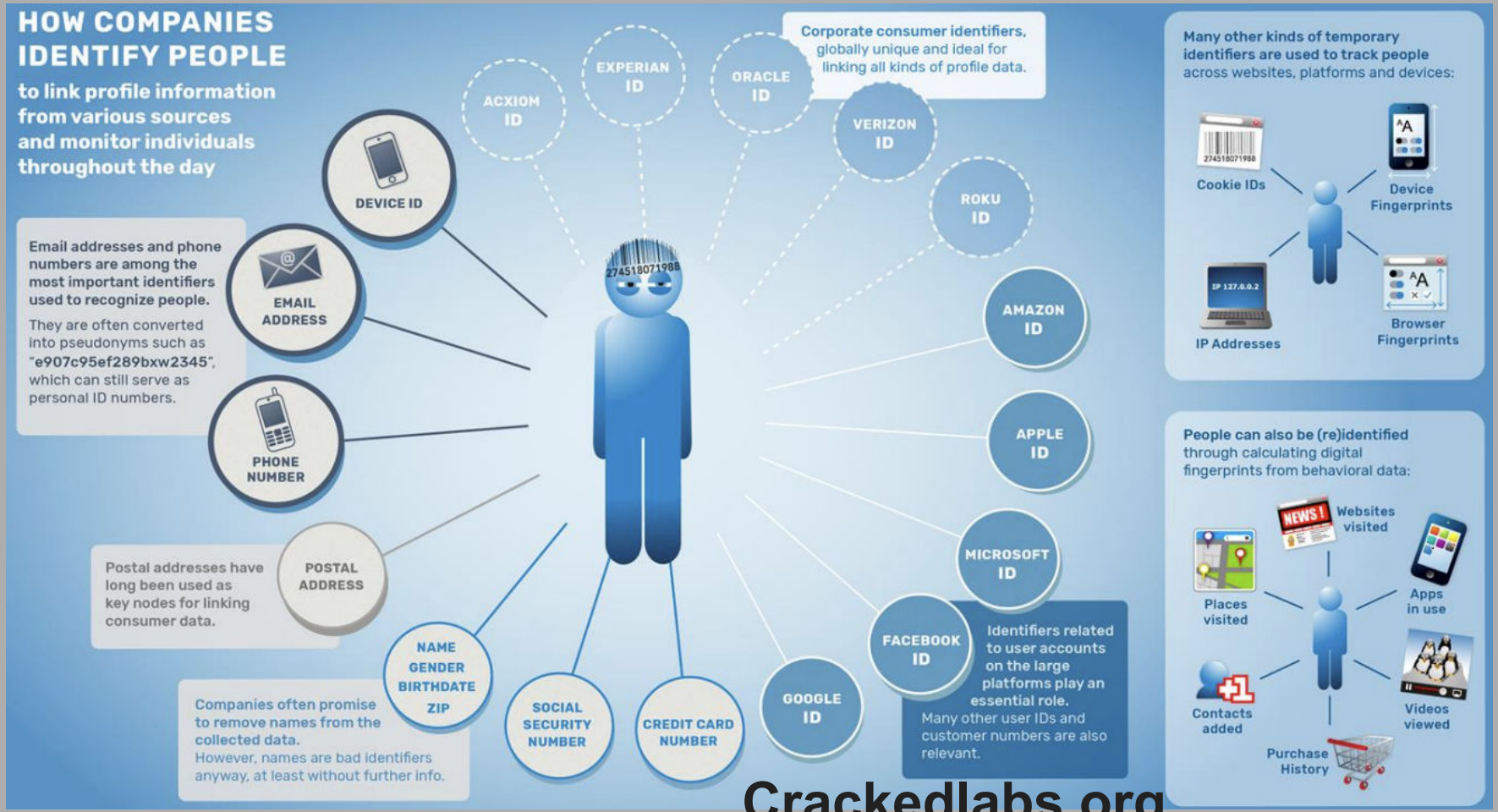
Server 2 Server



What is Server 2 Server (S2S)?



What is Server 2 Server (S2S)?



Crackedlabs.org

This data helps marketers:

Understand trend: spread, demographics, influence (nothing beats a jet2holiday)

Where: trends originate

What: accelerates/slows trends

And: who to target



How does this help marketers?

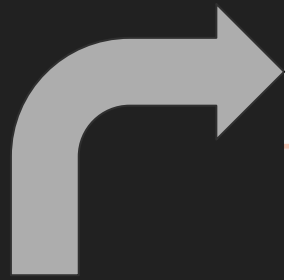
Two coworkers seem flirtatious? Push suggested restaurants, or activities from joint interests?



A couple are fighting a lot, push divorce lawyers, toxic masculinity products? Or maybe push Therapy services?



How does this help marketers?



These are *all* advantages?

Data Storage:	Stores data on a server
Implementation:	Complex to implement
Control:	Offers greater control over tracking and data management
Optimization:	Provides more optimization opportunities
Reliability:	More reliable
Tracking Duration:	Can track conversions indefinitely

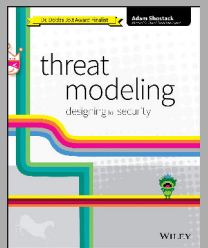


Privacy Harms

Information collection, information processing, information dissemination, invasion, and *identifier creation*

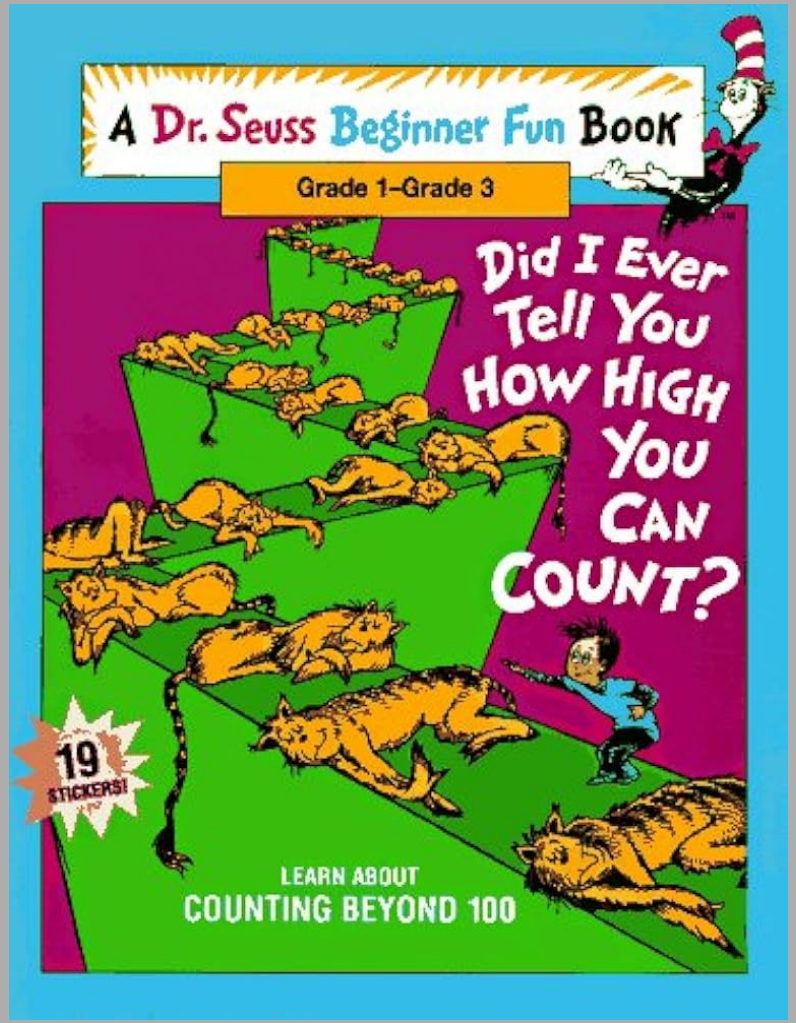
Enable: blackmail, increased accessibility, and distortion

Understanding Privacy, Professor Daniel Solove, Threat Modeling Designing for Security Adam Shostack



Server 2 Server goes much further than previous technologies

Cookies rely on towers to collect and tabulate data: Facebook, Google, etc



Privacy Harms

Towers see liability in holding data, often purposely sanitizing it, and limit external access

(yes yes, for greed based reasons)

S2S is the Wild West Lacking:

- *Standardization*
- *Centralization*
- *Regulation*
- *Sanitization*

Further Ethical Concerns



Further Ethical Concerns

Ads fuel misinformation, using emotional manipulation

Facebook was caught manipulating 690,000 users for one internal study - Kramer et al., 2014

Instagram leaks show they debated the morality of causing depression in tweens

WHAT ABOUT THE THREAT ACTOR YOUR PROMISED US!!!!!!



State Actors and Other Threats

In 2024 Russia spent $15.5 billion USD on advertising, only $1.9 of that was spent on Russian Audiences

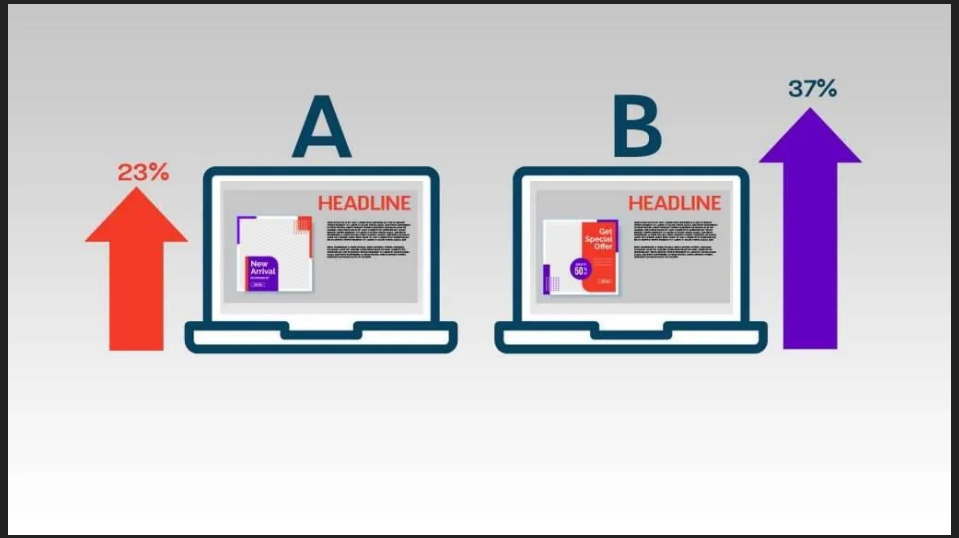
Promoting propaganda through ads, is kinda obvious right? Ads are labeled as such aren’t they?

Most of the adverts they’re placing are seemingly completely harmless and unrelated to the state actor

A/B Testing in Bot Farms

A common marketing tactic is to place many ads to gather up to the minute data on:

- Keywords
- Concepts
- Phrasings



The Internet's Dead Long Live the Internet

The Internet is dead.

The Bots have taken over.

What was once a theory is now a reality! The Internet is dead. Previously a bustling town hall of ideas has become an endless army of bots that gives the appearance of life, but is far from it



waking up to 100s of new followers that aren't even real! Well, you don't have to imagine anymore because the future is ushering in an Internet experience that is totally fake!

Micro Targeting

These ad nets, and purchased legitimate sets of data from ad data companies are powerful when combined

Data about age groups, hyper specific locations, and very focused interests is easy for threat actors to access

It allows them to sound just like someone in your peer group, making it difficult to tell if a frustrating conversation is with a human or not

It’s not Just Frustration

Bandwagoning, card stacking, transference

Are disinformation tactics pioneered in WWII

And perfected by the era of Mad Men

Have you ever shared more than you probably should to a peer on an industry forum?

Have you ever dismissed a threat because others didn’t seem to think it was a big deal?

Have you ever felt burnt out after a conversation on Reddit?

The Overlap

Threat actors and advertisers have the same goal

Increase awareness, align your opinion, drive an action

For advertisers, they just want you to buy something

For TA's it's something more malicious



Their strategies also overlap:

Having others naturally spread their messaging is the highest value they can achieve

Building datasets about households, and user interests increases efficacy

Cookies Never Die

Google starts a feud with their Frankenstein:

Ad agencies are unable to get GA4 to work

The maths not mathing

CTR is plummeting

Ad agencies directly call Google out:

Both of us will lose money

Stop being arrogant



AdAge

Google takes this poorly:

Taking out hit pieces in industry magazines

Ad agencies incompetent and greedy

CA4 gets delayed 4 times

January 2024 Google gave up

Turns out cookies are *safe and private!*



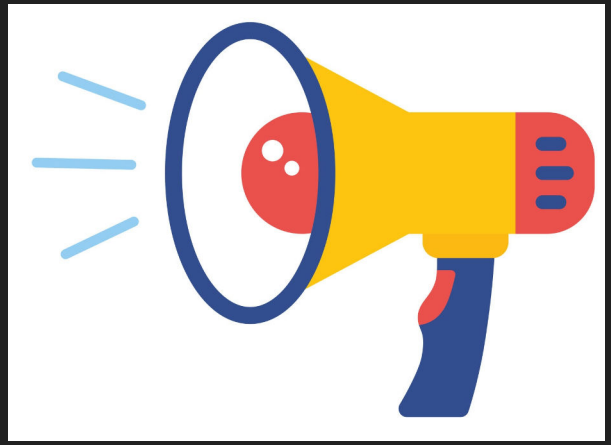


Cookies and S2S in harmony: Together producing more precise insights than ever before

Fostering partnerships: Google’s ReCaptcha, Meta Pixel

This victory doesn’t feel like *our* victory

Call to Action!



Outreach!

You can’t protect against the unknown

GDPR is severely underrated and highly effective

Call to Action!

Broad:

Likelihood of risk to rights and freedoms

Reasonable level of protection

Implementation of informed consent

Narrow:

Focused on data not harms

Obsessed with cookies

Legitimate interest

Anonymized but re-identifiable

Individual Recommendations

Privacy friendly browsers: FireFox, **not** Brave which was founded by a homophobe and funded by Theil

Isolated profiles: multi-account containers

VPNs don’t anonymize metadata, they can hide your SSID, geolocation, and more



Use ephemeral MAC addresses, Apple devices use them by default

Minimize IoT, segment networks, and use open source options when possible

Utilize a PiHole



Background

Sean “4dw@r3” Juroviesky

Founder of CyberPolyglots

Milliways Newbie

BurbSec Organizer



- siem.cyberpolyglots.org
- osint.cyberpolyglots.org
- lingua.cyberpolyglots.org
- Discord: https://discord.gg/qHGE5t5p6

Malware & Monsters

- malwareandmonsters.com
- Incident Response Tabletop Game themed off DnD
- Free Workshop Saturday at 12:00 meet at the CyberPolyglots booth





Thank You!

Questions?